



Effect of Integrated Nutrient Management on Production and Productivity of Garlic (*Allium sativum* L.) cv. Yamuna Safed

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Authors' contributions

This work was carried out in collaboration among all authors. Author AR designed the study, conducted the field experiment, performed the statistical analysis, and wrote the first draft of the manuscript. Author AS conceived and supervised the study, and reviewed and edited the manuscript. Author PN, MP and GG assisted with data collection, literature review, and manuscript revision. All authors read and approved the final manuscript.

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Abstract

Aims: To evaluate the influence of Integrated Nutrient Management (INM), combining inorganic fertilisers, farmyard manure (FYM), and biofertilisers (*Azotobacter* and phosphate-solubilising bacteria [PSB]), on the growth, yield attributes, bulb yield, and economic performance of garlic (*Allium sativum* L.) cv. Yamuna Safed, and to identify the most agronomically and economically viable nutrient combination for the Central Plateau and Hills Region of Satna, Madhya Pradesh.

Study Design: The field experiment was laid out in a randomised block design (RBD) with ten treatments (T0–T9), each replicated three times.

Place and Duration of Study: The study was conducted at the Horticulture Research Farm, AKS University, Satna, Madhya Pradesh, during the *Rabi* season of 2025–26.

Methodology: Cloves of cv. Yamuna Safed were planted on 04 November 2025 in plots measuring 2 m × 1 m. Treatments comprised a control, sole inorganic fertilisation (100% RDF), and graded combinations of reduced-dose RDF with FYM @ 5 t

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ha⁻¹, *Azotobacter* @ 2 kg ha⁻¹, and PSB @ 2 kg ha⁻¹. The RDF was 100:50:50 kg N:P:K ha⁻¹, row-to-row and plant-to-plant spacing was 15 cm × 10 cm, and the soil was light black. Growth parameters (plant height, leaves per plant, leaf length/width, and leaf area), yield attributes, bulb yield, and economic indices (cost of cultivation, gross and net returns, and benefit–cost ratio) were recorded and analysed statistically.

Results: Integrated application of nutrient sources significantly improved all growth, yield, and economic parameters compared with sole inorganic fertilisation. Treatment T8 (75% RDF + *Azotobacter* @ 2 kg ha⁻¹ + FYM @ 5 t ha⁻¹ + PSB @ 2 kg ha⁻¹) recorded the highest plant height (64.23 cm), number of leaves per plant (7.34), leaf area (48.84 cm²), bulb yield (152.32 q ha⁻¹), gross return (₹5,33,120 ha⁻¹), net return (₹4,27,120 ha⁻¹), and benefit–cost ratio (5.03).

Conclusion: Among the treatments evaluated, integrating 75% RDF with *Azotobacter* @ 2 kg ha⁻¹, FYM @ 5 t ha⁻¹, and PSB @ 2 kg ha⁻¹ was agronomically superior and economically the most viable nutrient-management strategy for garlic under conditions similar to those of the present study.

Keywords: Integrated nutrient management; garlic; Yamuna Safed; farmyard manure; *Azotobacter*; phosphate-solubilising bacteria; biofertilisers; bulb yield.

1. Introduction

Garlic (*Allium sativum* L.), a member of the family Amaryllidaceae, ranks among the most economically and therapeutically significant bulb crops cultivated globally. Its importance extends well beyond culinary use, as it has long been employed in traditional and modern medicine for its broad-spectrum therapeutic properties. The plant parts, including cloves, leaves, and inflorescences, have been used since antiquity as flavouring agents in soups, sausages, and salads (Verma *et al.*, 2023).

The pharmacological value of garlic is primarily attributed to its high concentration of organosulfur compounds, including allicin, diallyl disulphide, S-allylcysteine, and diallyl trisulfide, which together confer its characteristic medicinal properties. Garlic is consumed either fresh (raw or dried cloves) or as processed derivatives such as garlic oil, powder, and standardised extracts, each with a distinct bioactive profile. Historically, it has been used against intestinal disorders, respiratory infections, dermatological conditions, and bacterial diseases; the Babylonians, Egyptians, Phoenicians, Greeks, and Romans used it for worm infestations, wound care, and tumour suppression, and it was widely used against cholera, diphtheria, tuberculosis, and amoebic dysentery before the advent of antibiotics (Verma *et al.*, 2023).

India is the world's second-largest producer of garlic, with Madhya Pradesh contributing about 62% of national output and ranking as the leading garlic-producing state. The state's garlic production stands at approximately 2016 metric tonnes, and the crop is the third-largest spice by cultivated area. Predominant varieties grown in Madhya Pradesh include Yamuna Safed-3, Yamuna Safed-4 (G-323), Yamuna Safed-5 (G-189), Agrifound Parvati-2 (G-408), and G-282, together covering around 21% of the state's total spice area. Owing to its economic importance, garlic has been identified as a focus crop under state-sponsored agricultural market-linkage initiatives (Madhya Pradesh Trade Portal, 2026).

Recent field evaluation of diverse garlic genotypes has also demonstrated variation in bulb yield and related traits, supporting cultivar-specific interpretation of agronomic field performance (Fufa *et al.*, 2025).

Classified within the Liliaceae family, garlic is valued globally for its dual role as a medicinal plant and culinary ingredient. Its principal bioactive organosulfur compounds are rapidly absorbed and metabolised in the human body, giving it broad-spectrum therapeutic efficacy with minimal toxicity—a combination that continues to drive research interest in garlic cultivation and quality improvement (Verma *et al.*, 2023).

Integrated Nutrient Management (INM) is a holistic agronomic strategy that combines organic, inorganic, and biological nutrient sources to maintain soil fertility, optimise productivity, and minimise adverse environmental impacts (Thangasamy *et al.*, 2017). In garlic, INM has shown considerable potential to improve bulb size, yield, and quality (pungency and allicin content) while enhancing soil organic carbon, nutrient availability, and microbial activity. Combining organic amendments such as FYM, vermicompost, and poultry manure with biofertilisers (*Azotobacter* and PSB), alongside judiciously reduced doses of inorganic fertiliser, promotes sustained nutrient supply, improves nutrient-use efficiency, and reduces dependence on synthetic inputs (Thangasamy *et al.*, 2017). Such integrated approaches are increasingly advocated as a cornerstone of sustainable production, particularly where declining soil fertility and rising input costs challenge smallholder farmers. Despite this, region-specific information on the optimal INM combination for garlic cv. Yamuna Safed under the Kymore Plateau and Satpura Hills of Satna, Madhya Pradesh, remains limited; this knowledge gap formed the rationale for the present study.

Recent garlic studies further indicate that nutrient-management responses depend on the combination and source of applied nutrients, with effects reported for crop growth, bulb yield, quality, and nutrient-use performance (Arunachalam *et al.*, 2025; R. Choudhary *et al.*, 2024; B. Singh *et al.*, 2023).

1.1 Objective

The present study aimed to evaluate the effects of graded combinations of the recommended dose of fertiliser, farmyard manure, *Azotobacter*, phosphate-solubilising bacteria, and *Azospirillum* on the growth, yield attributes, bulb yield, and economic performance of garlic cv. Yamuna Safed under the agro-climatic conditions of Satna, Madhya Pradesh, and to identify the most agronomically and economically suitable nutrient combination.

2. Materials and Methods

2.1 Experimental Site

The field investigation was carried out during the *Rabi* season of 2025–26 at the Horticulture Research Farm, AKS University, Satna, Madhya Pradesh, India. Satna is located at approximately 24.34° N latitude and 80.49° E longitude, with an average elevation of about 315 metres (1,033 feet) above sea level. The site lies within the Kymore Plateau and Satpura Hills agro-climatic zone. The soil was light black, with pH 7.2, EC 0.19 dS/m, an organic carbon value of 0.83, and available N, P, and K contents of 176.5, 13.25, and 198.7 kg/ha, respectively.

2.2 Planting Material and Crop Establishment

Healthy, uniform cloves of garlic (*Allium sativum* L.) cv. Yamuna Safed were used as planting material and were planted on 04 November 2025. The planting material was treated with carbendazim and mancozeb before sowing. Row-to-row and plant-to-plant spacing was 15 cm × 10 cm, the seed rate was 500 kg/ha, and the planting material was purchased from Jawaharlal Nehru Krishi Vishwa Vidyalaya (JNKVV), Jabalpur (MP).

2.3 Experimental Design and Treatments

The experiment was laid out in a randomised block design (RBD) comprising ten treatments, each replicated three times, in individual plots measuring 2 m × 1 m. The recommended dose of fertiliser (RDF) was 100:50:50 kg N:P:K ha⁻¹. The treatment details were as follows:

- T0 – Absolute Control (no fertiliser/manure application)
- T1 – 100% RDF
- T2 – 50% RDF + *Azotobacter* @ 2 kg ha⁻¹
- T3 – 75% RDF + FYM @ 5 t ha⁻¹
- T4 – 50% RDF + PSB @ 2 kg ha⁻¹
- T5 – 25% RDF + *Azotobacter* @ 2 kg ha⁻¹ + FYM @ 5 t ha⁻¹
- T6 – 50% RDF + FYM @ 5 t ha⁻¹ + PSB @ 2 kg ha⁻¹
- T7 – 75% RDF + *Azotobacter* @ 2 kg ha⁻¹ + PSB @ 2 kg ha⁻¹
- T8 – 75% RDF + *Azotobacter* @ 2 kg ha⁻¹ + FYM @ 5 t ha⁻¹ + PSB @ 2 kg ha⁻¹
- T9 – *Azospirillum* @ 2 kg ha⁻¹ + FYM @ 5 t ha⁻¹ + PSB @ 2 kg ha⁻¹

FYM was incorporated into the soil 15 days before sowing, and *Azotobacter*, PSB, and *Azospirillum* biofertiliser cultures were applied according to the treatment details. The planting material was treated with carbendazim and mancozeb before sowing, and the biofertilisers were mixed into the soil as specified for each treatment. Inorganic fertilisers were applied by supplying 100% of the phosphorus (via SSP) and potassium (via MOP), together with 33%–50% of the nitrogen (via urea), as a basal dose during final land preparation. The remaining nitrogen was top-dressed in two split doses at 30 and 45–60 days after planting to maximise bulb bulking.

2.4 Crop Management

The standard recommended package of practices for garlic was followed uniformly across all treatments for irrigation, weeding, and plant protection. During the study, irrigation was applied on the third day after planting, followed by a 7–10-day irrigation interval during the vegetative and bulb-growth phases; irrigation was stopped two to three weeks before harvest. Two to three manual weeding were carried out during the first two months, and neem oil was used as an insecticide for pest control. The crop was harvested on 28 March upon attainment of physiological maturity, as indicated by the yellowing and drying of leaves.

2.5 Observations Recorded

Growth parameters—plant height (cm), number of leaves plant⁻¹, leaf length (cm), leaf width (cm), and leaf area (cm²)—were recorded from five randomly selected tagged plants per plot. Plant height was recorded at 30, 60, and 90 days after sowing; leaf length and leaf width were recorded at 60 and 90 days after sowing; and leaf area (cm²) was recorded at 60 and 90 days after sowing. Yield attributes recorded at harvest included number of cloves bulb⁻¹, average clove weight (g), yield plot⁻¹ (kg), and total bulb yield (q ha⁻¹), extrapolated on an area basis. Economic indices—cost of cultivation, gross return, net return, and benefit–cost (B:C) ratio—were computed using prevailing market prices of inputs and produce during the study period. Gross return was calculated by multiplying bulb yield by the prevailing market price; net return was calculated as gross return minus the cost of cultivation; and the B:C ratio was calculated as gross return divided by the cost of cultivation.

2.6 Statistical Analysis

Data recorded for each parameter were statistically analysed using the analysis of variance (ANOVA) technique applicable to a randomised block design (Panse & Sukhatme, 1967). Treatment means were compared using the Critical Difference (CD) at the 5% probability level ($P = 0.05$), and the standard error of the mean (S.E.m. ±) was computed for each parameter. The data were analysed

using OPSTAT, an online agricultural statistical analysis software package provided by Chaudhary Charan Singh Haryana Agricultural University (CCS HAU), Hisar, Haryana, India.

3. Results and Discussion

3.1 Growth Parameters

Table 1. Effect of Integrated Nutrient Management (INM) on growth parameters of garlic

Treatment	Plant Height (cm)	No. of Leaves Plant ⁻¹	Leaf Length (cm)	Leaf Width (cm)	Leaf Area (cm ²)
T0 – Absolute Control (no fertiliser/manure application)	51.89	5.12	36.47	2.85	37.39
T1 – 100% RDF	53.80	5.56	38.75	2.99	38.90
T2 – 50% RDF + Azotobacter @ 2 kg ha ⁻¹	59.33	6.35	39.26	3.15	39.66
T3 – 75% RDF + FYM @ 5 t ha ⁻¹	59.93	6.24	40.69	3.31	40.31
T4 – 50% RDF + PSB @ 2 kg ha ⁻¹	60.38	6.33	42.91	3.49	41.77
T5 – 25% RDF + Azotobacter @ 2 kg ha ⁻¹ + FYM @ 5 t ha ⁻¹	60.97	6.77	44.81	3.82	43.30
T6 – 50% RDF + FYM @ 5 t ha ⁻¹ + PSB @ 2 kg ha ⁻¹	62.35	6.66	46.32	4.00	45.65
T7 – 75% RDF + Azotobacter @ 2 kg ha ⁻¹ + PSB @ 2 kg ha ⁻¹	63.04	7.24	46.97	4.22	46.31
T8 – 75% RDF + Azotobacter @ 2 kg ha ⁻¹ + FYM @ 5 t ha ⁻¹ + PSB @ 2 kg ha ⁻¹	64.23	7.34	48.82	4.41	48.84
T9 – Azospirillum @ 2 kg ha ⁻¹ + FYM @ 5 t ha ⁻¹ + PSB @ 2 kg ha ⁻¹	61.46	6.70	43.19	3.64	42.65
S.Em ±	0.91	0.25	1.06	0.17	0.73
CD (P = 0.05)	2.66	0.73	3.09	0.49	2.14

Integrated nutrient management exerted a statistically significant influence on the vegetative growth attributes of garlic (Table 1). Among all treatments, T8 (75% RDF + *Azotobacter* @ 2 kg ha⁻¹ + FYM @ 5 t ha⁻¹ + PSB @ 2 kg ha⁻¹) consistently recorded the maximum plant height, number of leaves per plant, leaf length, leaf width, and leaf area. The superior vegetative performance under this treatment may be attributed to the synergistic interaction among the inorganic fertiliser component, organic manure, and microbial inoculants, which together provided a balanced and sustained nutrient supply throughout the crop cycle.

Azotobacter contributed to atmospheric nitrogen fixation, augmenting the nitrogen pool available to the crop, while PSB facilitated the solubilisation of fixed soil phosphorus into plant-available forms. FYM, in turn, improved soil aggregation, porosity, and moisture retention and stimulated beneficial rhizosphere microbial activity, thereby enhancing root proliferation and nutrient acquisition (Kumari *et al.*, 2024; A. K. Singh *et al.*, 2023; Nainwal *et al.*, 2015; Yadav & Pandey, 2025). The resulting improvement in macro- and micronutrient availability promoted cell division, cell elongation, and chlorophyll biosynthesis, culminating in robust vegetative growth. These observations are consistent with previous reports of comparable improvements in garlic growth under integrated nutrient regimes (Choudhary *et al.*, 2013; Eragegowda *et al.*, 2018; Damse *et al.*, 2014; Meena *et al.*, 2019; Priyanshu *et al.*, 2019).

3.2 Yield Attributes and Bulb Yield

Integrated nutrient management also produced marked improvements in yield-attributing traits and overall bulb productivity (Table 2). Treatment T8 (75% RDF + *Azotobacter* @ 2 kg ha⁻¹ + FYM @ 5 t ha⁻¹ + PSB @ 2 kg ha⁻¹) registered the highest number of cloves per bulb, individual clove weight, plot-level yield, and extrapolated bulb yield per hectare. This yield enhancement can be mechanistically linked to the superior vegetative growth and greater leaf area observed under this treatment, which likely enhanced photosynthetic efficiency and the translocation of photo-assimilates towards developing bulbs (Thangasamy & Lawande, 2018; Pathak & Shukla, 2025; V. Singh *et al.*, 2017b; Priyanshu *et al.*, 2020; Bhandari *et al.*, 2012). The combined application of reduced-dose chemical fertiliser, FYM, *Azotobacter*, and PSB provided a favourable nutrient-release profile, with immediate availability from inorganic sources complemented by slow-release mineralisation from organic manure and biologically fixed nutrients, thereby ensuring uninterrupted nutrition during bulb initiation and bulking. These results are consistent with reports of enhanced bulb yield in garlic under integrated nutrient management (A. K. Singh *et al.*, 2023; Chouhan *et al.*, 2026; Choudhary & Kumari, 2025; Garg *et al.*, 2025; Patidar *et al.*, 2017).

3.3 Economic Analysis

The economic assessment demonstrated the superior financial viability of integrated nutrient management over sole chemical fertilisation (Table 2). Treatment T8 (75% RDF + *Azotobacter* @ 2 kg ha⁻¹ + FYM @ 5 t ha⁻¹ + PSB @ 2 kg ha⁻¹) recorded the highest gross return (₹5,33,120 ha⁻¹), net return (₹4,27,120 ha⁻¹), and benefit–cost ratio (5.03). Although the cultivation cost under this treatment was marginally higher because of FYM and biofertiliser inputs, the substantially higher bulb yield more than offset this additional expenditure. In contrast, the absolute control (T0) recorded the poorest economic performance, attributable to inadequate nutrient supply and consequently reduced growth and yield. These findings indicate that integrated nutrient management enhances not only agronomic productivity but also farm profitability, in agreement with previous reports (V. Singh *et al.*, 2017a; Magray *et al.*, 2017; Kumar & Kamboj, 2019; Patel *et al.*, 2019; Tripathy *et al.*, 2017).

Table 2. Effect of integrated nutrient management (INM) on yield and economics of garlic

Treatment	Cloves Bulb ⁻¹	Clove Weight (g)	Yield (kg) Plot ⁻¹	Total Yield (q ha ⁻¹)	Cost of Cultivation (Rs ha ⁻¹)	Gross Return (Rs ha ⁻¹)	Net Return (Rs ha ⁻¹)	B:C Ratio
T0 – Absolute Control (no fertiliser/manure application)	14.52	1.06	5.82	110.57	85,000	3,86,995	3,01,995	4.55
T1 – 100% RDF	15.35	1.14	6.18	113.25	86,000	3,96,375	3,10,375	4.61
T2 – 50% RDF + Azotobacter @ 2 kg ha ⁻¹	16.51	1.21	6.52	116.02	86,500	4,06,070	3,19,570	4.69
T3 – 75% RDF + FYM @ 5 t ha ⁻¹	19.81	1.27	6.92	120.73	90,000	4,22,555	3,32,555	4.70
T4 – 50% RDF + PSB @ 2 kg ha ⁻¹	20.97	1.34	7.12	123.02	91,000	4,30,570	3,39,570	4.73
T5 – 25% RDF + Azotobacter @ 2 kg ha ⁻¹ + FYM @ 5 t ha ⁻¹	24.07	1.48	7.68	138.66	1,02,000	4,85,310	3,83,310	4.76
T6 – 50% RDF + FYM @ 5 t ha ⁻¹ + PSB @ 2 kg ha ⁻¹	25.75	1.59	7.98	143.47	1,03,000	5,02,145	3,99,145	4.88
T7 – 75% RDF + Azotobacter @ 2 kg ha ⁻¹ + PSB @ 2 kg ha ⁻¹	27.22	1.67	8.20	149.00	1,04,500	5,21,500	4,17,000	4.99
T8 – 75% RDF + Azotobacter @ 2 kg ha ⁻¹ + FYM @ 5 t ha ⁻¹ + PSB @ 2 kg ha ⁻¹	29.08	1.80	8.48	152.32	1,06,000	5,33,120	4,27,120	5.03
T9 – Azospirillum @ 2 kg ha ⁻¹ + FYM @ 5 t ha ⁻¹ + PSB @ 2 kg ha ⁻¹	22.94	1.46	7.38	127.46	94,100	4,46,110	3,52,010	4.74
S.Em ±	1.40	0.09	0.25	1.58	–	–	–	–
CD (P = 0.05)	4.08	0.26	0.74	4.61	–	–	–	–

4. Conclusion

The present investigation indicates that integrated nutrient management significantly enhanced the growth parameters, yield attributes, and economic returns of garlic cv. Yamuna Safed under the agro-climatic conditions of Satna, Madhya Pradesh. The combined application of inorganic fertilisers, FYM, and biofertilisers was more effective than sole chemical fertilisation in improving overall crop performance. Among the treatments evaluated, T8 (75% RDF + *Azotobacter* @ 2 kg ha⁻¹ + FYM @ 5 t ha⁻¹ + PSB @ 2 kg ha⁻¹) recorded the highest values across all growth and yield parameters, together with the maximum gross return, net return, and benefit–cost ratio. Based on these results, this combination may be considered an economically viable and ecologically sustainable nutrient-management strategy for garlic cultivation under agro-climatic conditions similar to those of the present study. Further multi-location and multi-season trials are suggested to validate these findings before broader recommendations are made.

5. Limitation

The study was conducted during a single *Rabi* season at one experimental location in Satna, Madhya Pradesh, using one garlic cultivar, Yamuna Safed. Therefore, the observed treatment responses may be specific to the prevailing soil, climate, management conditions, and season. The economic analysis was also based on input and produce prices prevailing during the study period. Multi-season and multi-location evaluation would be useful to assess the consistency of the agronomic and economic responses before wider recommendation.

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Declaration of AI Use

This manuscript was prepared through the intellectual contributions of the author(s) to the study design, data analysis, interpretation of results, and scholarly content. Grammarly and ChatGPT were used solely to assist with grammatical refinement and reference formatting during manuscript revision. The author(s) accept full responsibility for the accuracy, integrity, and final content of the manuscript.

Competing Interests

Authors have declared that no competing interests exist.

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